Robust Spanning Trees under Discrete and Interval Uncertainty

Min-Max and Min-Max Regret Models

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[Date of defence]

Final Presentation of the Bachelor Thesis in Computer Science
Combinatorial Optimization, Prof. Dr. Christina Büsing, RWTH Aachen University

Spanning Trees under Uncertainty

Committing to a Tree before Costs Are Known

- Minimum spanning tree: connect all vertices at minimum total cost
- Classical setting: costs known, solved greedily in $O(m \log n)$

[TikZ: network with uncertain costs]

Committing to a Tree before Costs Are Known

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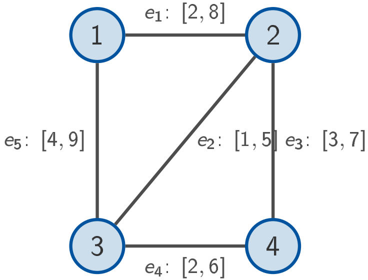
- Classical setting: costs known, solved greedily in $O(m \log n)$
- This talk: the tree is fixed first, the costs arrive later

[TikZ: network with uncertain costs]

One Example for the Whole Talk



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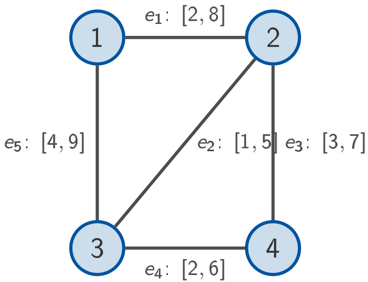

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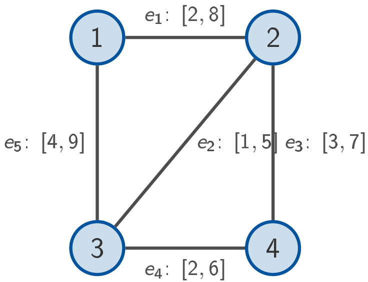

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- Scenarios: $c^{(1)} = (2, 1, 3, 2, 4)$,
  $c^{(2)} = (8, 5, 7, 6, 9)$,
  $c^{(3)} = (5, 1, 7, 2, 4)$

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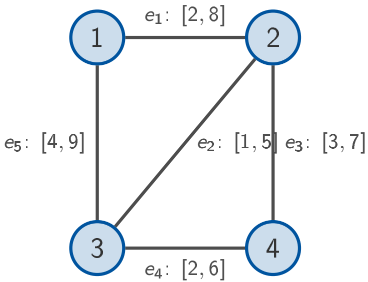

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  $c^{(2)} = (8, 5, 7, 6, 9)$,
  $c^{(3)} = (5, 1, 7, 2, 4)$
- Trees to watch:
  $T_1 = \{e_1, e_2, e_3\}$,
  $T_2 = \{e_1, e_2, e_4\}$,
  $T_3 = \{e_2, e_3, e_5\}$

Three Adversaries

Definition (uncertainty sets)

- Discrete: $\mathcal{U} = \{c^{(1)}, \ldots, c^{(K)}\}$, one full scenario is chosen
- Interval: $\mathcal{U} = \prod_{e \in E} [\ell_e, u_e]$, every edge varies independently
- Budgeted: intervals, but at most Γ edges may deviate upwards

Two Robust Objectives

Min-max

$$\min_{T \in \mathcal{T}} \max_{c \in \mathcal{U}} c(T)$$

Min-max regret

$$\min_{T \in \mathcal{T}} \max_{c \in \mathcal{U}} \left(c(T) - \mathrm{MST}(c) \right)$$

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- On the micro-graph the two objectives pick different trees

Min-Max Spanning Trees

Intervals: the Adversary Has No Real Choice

Lemma

For every spanning tree T it holds that $\mathrm{wc}(T) = \sum_{e \in E(T)} u_e$.

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Source: thesis Lemma 3.1; cf. Goerigk, Theorem 4.8.

Discrete Scenarios: Two Are Already Hard

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- Reduction from the *Partition* problem

Source: thesis Theorem 3.2; cf. Goerigk, Theorem 8.4.

The Reduction, Drawn (I)

[TikZ: partition gadget, construction clicks]

The Reduction, Drawn (II)

[TikZ: tree choice $\leftrightarrow$ partition, equivalence clicks]

The Complete Min-Max Picture

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Budgeted	polynomial	$O(m)$ MST computations

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Source: thesis §3.4 (in progress); cf. Goerigk, Theorem 4.21.

Min-Max Regret Spanning Trees

Regret: Competing with Hindsight

[Blocked: mirrors thesis Chapter 4]

Regret under Interval Uncertainty

[Blocked: mirrors thesis Chapter 4]

Summary

- One tree, adversarial costs: difficulty is set by the uncertainty model
- Intervals polynomial, two scenarios already NP-hard, a deviation budget restores tractability
- Regret: [finalise with Ch4]

Open questions

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Thank you for your attention!

References

[Bibliography of slide-cited sources: Goerigk, Korte-Vygen, . . .]

Backup: Fundamental Cut Lemma

Backup: FPTAS for Constant K

Backup: Budgeted Worked Example on the Micro-Graph

Backup: Full Micro-Graph Cost Table